Lower bound reasoning with eHL

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Notation. For a predicate P we use two indicator-style operators, one tailored to each direction of reasoning:

$$[P] \;=\; \begin{cases} 1 & \text{if } P \\ 0 & \text{otherwise} \end{cases} \qquad \text{(used for lower bounds)}$$

$$\langle P \rangle \;=\; \begin{cases} 1 & \text{if } P \\ \infty & \text{otherwise} \end{cases} \qquad \text{(used for upper bounds)}$$

Given a (state-indexed) expectation r, the expressions $[P] \cdot r$ and $\langle P \rangle \cdot r$ denote the corresponding pointwise products: $[P] \cdot r$ agrees with r on states satisfying P and is zero elsewhere, while $\langle P \rangle \cdot r$ agrees with r on states satisfying P and is ∞ elsewhere.

The witness sequence $\{r_i\}_{i \in \mathbb{N}}$. The rule of Figure 2 uses a sequence $\{r_i\}_{i \geq 0}$ of non-negative expectations as a termination witness: intuitively, r_i is a (user-chosen) upper bound on how much of the post-expectation mass of f is still trapped inside the loop after i body executions. Premise 2 ties r_i to f at $i = 0$ by requiring $r_0 \geq f$; Premise 3 propagates the bound across one iteration of the body; and Premise 4 forces the bound — and thus the trapped mass — to vanish in the limit.

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$$\frac{}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{skip}\ \{\, P \mid f \,\}}\ [\textsc{skip}]$$

$$\frac{}{\{\, P[\mathsf{x}/\mathsf{E}] \mid f[\mathsf{x}/\mathsf{E}] \,\}_{\bowtie}\ \mathsf{x} \leftarrow \mathsf{E}\ \{\, P \mid f \,\}}\ [\textsc{assign}]$$

$$\frac{}{\{\, \forall v \in \mathsf{supp}(\mathsf{S}),\ P[\mathsf{x}/v] \mid \mathbb{E}_{\mathsf{S}}[\lambda v.\, f[\mathsf{x}/v]] \,\}_{\bowtie}\ \mathsf{x} \xleftarrow{\$} \mathsf{S}\ \{\, P \mid f \,\}}\ [\textsc{sample}]$$

$$\frac{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, R \mid h \,\} \quad \{\, R \mid h \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C};\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{seq}]$$

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, Q \mid g \,\} \quad \{\, \neg\mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\ \texttt{else}\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{if}]$$

$$\frac{\mathsf{B} \wedge P \vDash f_0 \bowtie 0 \quad \forall k.\, \{\, \mathsf{B} \wedge P \mid f_{k+1} \,\}_{\bowtie}\ \mathsf{C}\ \{\, P \mid f_k \,\} \quad \neg\mathsf{B} \wedge P \vDash f_k \bowtie g}{\{\, P \mid \mathsf{lub} f_k \,\}_{\bowtie}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid g \,\}}\ [\textsc{while}]$$

Fig. 1. eHL for lower and upper-bounds. Here $\bowtie \in \{\leq, \geq\}$.

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\leq}\ \mathsf{C}\ \{\, P \mid f \,\} \quad \mathsf{B} \wedge P \vDash r_0 \geq f \quad \forall i \geq 0.\, \{\, \mathsf{B} \wedge P \mid r_{i+1} \,\}_{\geq}\ \mathsf{C}\ \{\, \top \mid [P] \cdot r_i \,\} \quad \mathsf{B} \wedge P \vDash \lim_{i\to\infty} r_i = 0}{\{\, P \mid f \,\}_{\leq}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid f \,\}}\ [\textsc{wh}$$

Fig. 2. The While rule for eHL lower bounds. The rule is specific to the lower-bound judgment: Premise 1 is a lower-bound triple on the body, while Premise 3 is an *upper*-bound triple used purely to bound the in-loop mass. The sequence $\{r_i\}_{i\geq 0}$ consists of non-negative expectations $r_i : \mathsf{State} \to \mathbb{R}^+$.

A Soundness of the While Rule

Goal. We want to show that the following rule is sound:

$$[P] \cdot f \ \leq\ \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg\mathsf{B}] \cdot f)$$

Definitions. Define the family of approximating programs:

$$W_n \ \triangleq\ \big(\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\big)^n\ ;\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \texttt{abort}$$

where W_n allows at most n iterations; if the loop would continue beyond n, it aborts. Define:

$$I_n \ \triangleq\ [P] \cdot \mathsf{wp}\,[W_n]\,([P \wedge \neg\mathsf{B}] \cdot f)$$

Lemma A.1 (Recursive characterisation of I_n).

$$I_0 \ =\ [P \wedge \neg\mathsf{B}] \cdot f \qquad \textit{and} \qquad I_{n+1} \ =\ [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg\mathsf{B}] \cdot f$$

Proof. By induction on n.

Base case $n = 0$.

$$\begin{aligned} I_0 &= [P] \cdot \mathsf{wp}\,[\texttt{if}\ \mathsf{B}\ \texttt{then}\ \texttt{abort}]\,([P \wedge \neg\mathsf{B}] \cdot f) \\ &= [P] \cdot \Big([\mathsf{B}] \cdot \mathsf{wp}\,[\texttt{abort}]\,([P \wedge \neg\mathsf{B}] \cdot f) + [\neg\mathsf{B}] \cdot [P \wedge \neg\mathsf{B}] \cdot f\Big) \\ &= [P] \cdot \big([\mathsf{B}] \cdot 0 + [\neg\mathsf{B}] \cdot [P \wedge \neg\mathsf{B}] \cdot f\big) \\ &= [P] \cdot [P \wedge \neg\mathsf{B}] \cdot f \ =\ [P \wedge \neg\mathsf{B}] \cdot f \quad \checkmark \end{aligned}$$

where we used $\mathsf{wp}\,[\texttt{abort}]\,(\cdot) = 0$ and $[P] \cdot [P \wedge \neg\mathsf{B}] = [P \wedge \neg\mathsf{B}]$.

Inductive step $n \to n+1$.

$$I_{n+1} = [P] \cdot \mathsf{wp}\left[(\texttt{if B then C})^{n+1}\ ;\ \texttt{if B then abort}\right]([P \wedge \neg \mathsf{B}] \cdot f)$$

Unfolding $(\texttt{if B then C})^{n+1} = \texttt{if B then C}\ ;\ (\texttt{if B then C})^{n}$ and applying the SEQ rule:

$$= [P] \cdot \mathsf{wp}\,[\texttt{if B then C}]\,(\mathsf{wp}\,[W_n]\,([P \wedge \neg \mathsf{B}] \cdot f))$$

By sub-induction one verifies $[P] \cdot I_k = I_k$ for all k (since $I_0 = [P \wedge \neg \mathsf{B}] \cdot f$ and each I_{k+1} carries $[P \wedge \mathsf{B}]$ or $[P \wedge \neg \mathsf{B}]$ as a factor), so in particular:

$$\mathsf{wp}\,[W_n]\,([P \wedge \neg \mathsf{B}] \cdot f) \;=\; [P] \cdot \mathsf{wp}\,[W_n]\,([P \wedge \neg \mathsf{B}] \cdot f) \;=\; I_n$$

Therefore:

$$\begin{aligned} &= [P] \cdot \mathsf{wp}\,[\texttt{if B then C}]\,(I_n) \\ &= [P] \cdot \big([\mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_n) + [\neg \mathsf{B}] \cdot I_n\big) \\ &= [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_n) + [P \wedge \neg \mathsf{B}] \cdot I_n \end{aligned}$$

Using $[P] \cdot I_n = I_n$ and the identity $[P \wedge \neg \mathsf{B}] \cdot I_n = [P \wedge \neg \mathsf{B}] \cdot f$ (which follows from $[\neg \mathsf{B}] \cdot I_n = [\neg \mathsf{B}] \cdot [P \wedge \neg \mathsf{B}] \cdot f = [P \wedge \neg \mathsf{B}] \cdot f$):

$$I_{n+1} \;=\; [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg \mathsf{B}] \cdot f \quad \checkmark$$

□

Lemma A.2 (Key inequality). *Assuming premises* (a), (b), (c), *for all* $n \geq 0$*:*

$$[P] \cdot I_n + [P] \cdot r_n \;\geq\; [P] \cdot f$$

Proof. By induction on n.

Base case $n = 0$.

$$[P] \cdot I_0 + [P] \cdot r_0 \;=\; [P \wedge \neg \mathsf{B}] \cdot f + [P] \cdot r_0$$

By premise (b), $[P] \cdot r_0 \geq [P \wedge \mathsf{B}] \cdot f$, so:

$$\geq\; [P \wedge \neg \mathsf{B}] \cdot f + [P \wedge \mathsf{B}] \cdot f \;=\; [P] \cdot f \quad \checkmark$$

Inductive step $n \to n+1$. Expanding I_{n+1} by Theorem A.1:

$$[P] \cdot I_{n+1} + [P] \cdot r_{n+1} = [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg \mathsf{B}] \cdot f + [P] \cdot r_{n+1}$$

By premise (c), $[P] \cdot r_{n+1} \geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n)$:

$$\geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n) + [P \wedge \neg \mathsf{B}] \cdot f$$

By linearity of $\mathsf{wp}\,[\mathsf{C}]\,(\cdot)$:

$$= [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n + [P] \cdot r_n) + [P \wedge \neg \mathsf{B}] \cdot f$$

By the induction hypothesis $[P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ and monotonicity of $\mathsf{wp}\,[\mathsf{C}]\,(\cdot)$:

$$\geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f) + [P \wedge \neg \mathsf{B}] \cdot f$$

By premise (a), $[P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f) \geq [P \wedge \mathsf{B}] \cdot f$:

$$\geq\; [P \wedge \mathsf{B}] \cdot f + [P \wedge \neg \mathsf{B}] \cdot f \;=\; [P] \cdot f \quad \checkmark$$

□

Main result. From Theorem A.2, for all $n \geq 0$:

$$[P] \cdot I_n + [P] \cdot r_n \;\geq\; [P] \cdot f$$

Taking the limit:

$$\lim_n([P] \cdot I_n) + \lim_n([P] \cdot r_n) \;\geq\; [P] \cdot f$$

By condition (d), $\lim_n r_n = 0$, so $\lim_n([P] \cdot r_n) = 0$:

$$\lim_n([P] \cdot I_n) \;\geq\; [P] \cdot f$$

By Scott-continuity and Theorem A.1, $\lim_n I_n = \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg\mathsf{B}] \cdot f)$, giving:

$$\mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg\mathsf{B}] \cdot f) \;\geq\; [P] \cdot f$$

which is exactly the desired conclusion. □

B Relative Completeness

Theorem: Relative Completeness. **Assumptions:**

1. $[P] \cdot f \;\leq\; \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg\mathsf{B} \wedge P] \cdot f)$
2. $[\mathsf{B} \wedge P] \cdot f \;\leq\; \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f)$

Definitions:

$$\begin{aligned} H &\triangleq \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg\mathsf{B} \wedge P] \cdot f) \\ I_0 &\triangleq [\neg\mathsf{B} \wedge P] \cdot f \\ I_{i+1} &\triangleq [\mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_i) + I_0 \\ r_i &\triangleq H - I_i \end{aligned}$$

Verification of Premise 2:

$$\begin{aligned} \mathsf{B} \wedge P \implies I_0 &= 0 \\ r_0 = H - I_0 &= H \\ \text{From (1): } H \geq [P] \cdot f &\implies H \geq f \\ [\mathsf{B} \wedge P] \cdot r_0 &\geq [\mathsf{B} \wedge P] \cdot f \end{aligned}$$

Verification of Premise 2:

$\mathsf{B} \wedge P \implies I_0 = 0$

$r_0 = H - I_0 = H$

P holds so: $H \geq [P] \cdot f \implies H \geq f$

$[\mathsf{B} \wedge P] \cdot r_0 \geq [\mathsf{B} \wedge P] \cdot f$

Verification of Premise 3:

$$
\begin{aligned}
H &= [\mathsf{B}] \cdot \mathsf{wp}\ [\mathsf{C}]\ (H) + I_0 \\
r_{i+1} &= H - I_{i+1} \\
r_{i+1} &= ([\mathsf{B}] \cdot \mathsf{wp}\ [\mathsf{C}]\ (H) + I_0) - ([\mathsf{B}] \cdot \mathsf{wp}\ [\mathsf{C}]\ (I_i) + I_0) \\
r_{i+1} &= [\mathsf{B}] \cdot (\mathsf{wp}\ [\mathsf{C}]\ (H) - \mathsf{wp}\ [\mathsf{C}]\ (I_i)) \\
r_{i+1} &= [\mathsf{B}] \cdot \mathsf{wp}\ [\mathsf{C}]\ (H - I_i) = [\mathsf{B}] \cdot \mathsf{wp}\ [\mathsf{C}]\ (r_i) \\
\text{By Monotonicity: } r_i &\geq [P] \cdot r_i \implies \mathsf{wp}\ [\mathsf{C}]\ (r_i) \geq \mathsf{wp}\ [\mathsf{C}]\ ([P] \cdot r_i) \\
[\mathsf{B} \wedge P] \cdot r_{i+1} &\geq [\mathsf{B} \wedge P] \cdot \mathsf{wp}\ [\mathsf{C}]\ ([P] \cdot r_i)
\end{aligned}
$$

Verification of Premise 4:

$$
\begin{aligned}
&\lim_{i \to \infty} I_i = H \\
&\lim_{i \to \infty} r_i = H - \lim_{i \to \infty} I_i = H - H = 0 \\
&\lim_{i \to \infty} r_i = 0
\end{aligned}
$$